



Анализ и визуализация данных по трафику сетевого устройства, используя Netflow протокол и БД Clickhouse

ClickHouse для инженеров и архитекторов БД

План доклада

Общая схема конвейера данных

Настройка mikrotik как источник Netflow

Настройка goflow

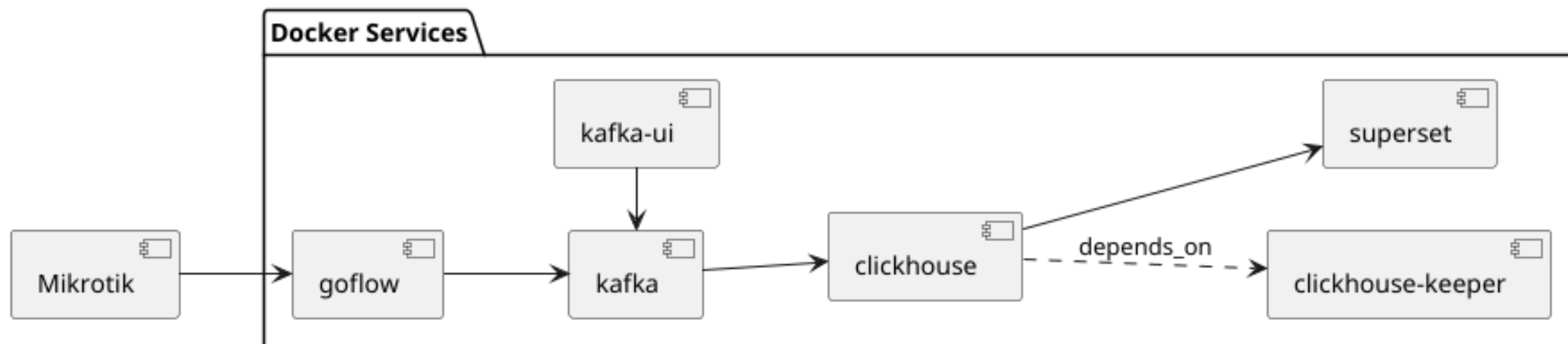
Настройка Kafka

Настройка Clickhouse

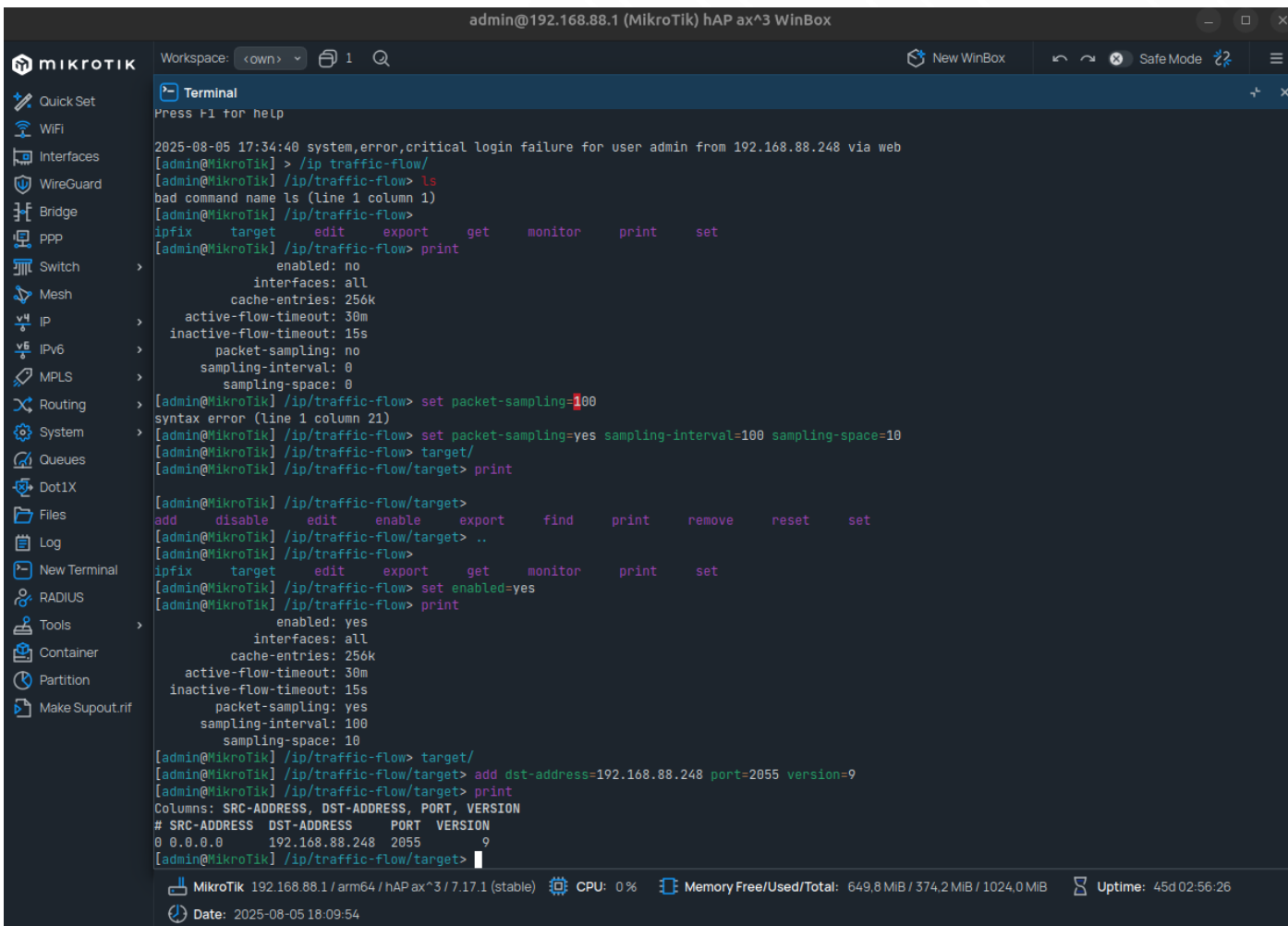
Настройка Apache Superset

Заключение

Общая схема конвейера данных



Настройка mikrotik hAP ax^3 как источник данных Netflow



The screenshot shows the Mikrotik WinBox interface with a terminal window open. The terminal displays the following commands and their outputs:

```
admin@192.168.88.1 (MikroTik) hAP ax^3 WinBox
Workspace: <own> 1
Terminal
Press F1 for help
2025-08-05 17:34:40 system,error,critical login failure for user admin from 192.168.88.248 via web
[admin@MikroTik] > /ip traffic-flow/
[admin@MikroTik] /ip/traffic-flow> ls
bad command name ls (line 1 column 1)
[admin@MikroTik] /ip/traffic-flow>
ipfix      target      edit      export    get      monitor   print     set
[admin@MikroTik] /ip/traffic-flow> print
enabled: no
interfaces: all
cache-entries: 256k
active-flow-timeout: 30m
inactive-flow-timeout: 15s
packet-sampling: no
sampling-interval: 0
sampling-space: 0
[admin@MikroTik] /ip/traffic-flow> set packet-sampling=100
syntax error (line 1 column 21)
[admin@MikroTik] /ip/traffic-flow> set packet-sampling=yes sampling-interval=100 sampling-space=10
[admin@MikroTik] /ip/traffic-flow> target/
[admin@MikroTik] /ip/traffic-flow/target> print
add  disable  edit  enable  export  find  print  remove  reset  set
[admin@MikroTik] /ip/traffic-flow/target> ..
[admin@MikroTik] /ip/traffic-flow>
ipfix      target      edit      export    get      monitor   print     set
[admin@MikroTik] /ip/traffic-flow> set enabled=yes
[admin@MikroTik] /ip/traffic-flow> print
enabled: yes
interfaces: all
cache-entries: 256k
active-flow-timeout: 30m
inactive-flow-timeout: 15s
packet-sampling: yes
sampling-interval: 100
sampling-space: 10
[admin@MikroTik] /ip/traffic-flow> target/
[admin@MikroTik] /ip/traffic-flow/target> add dst-address=192.168.88.248 port=2055 version=9
[admin@MikroTik] /ip/traffic-flow/target> print
Columns: SRC-ADDRESS, DST-ADDRESS, PORT, VERSION
# SRC-ADDRESS  DST-ADDRESS  PORT  VERSION
0 0.0.0.0      192.168.88.248 2055   9
[admin@MikroTik] /ip/traffic-flow/target>
```

The status bar at the bottom shows: MikroTik 192.168.88.1 / arm64 / hAP ax^3 / 7.17.1 (stable) CPU: 0% Memory Free/Used/Total: 649,8 MiB / 374,2 MiB / 1024,0 MiB Uptime: 45d 02:56:26 Date: 2025-08-05 18:09:54

Для настройки использовалась стандартная инструкция от Mikrotik

5 Настройка goflow

```
goflow:  
image: cloudflare/goflow:latest  
container_name: goflow  
restart: unless-stopped  
entrypoint: '/bin/sh'  
command: >  
-c "sleep 15 ;  
/goflow -kafka.brokers kafka:9092  
-kafka.topic flows  
-proto.fixedlen=true  
"  
  
ports:  
- '8081:8080'  
- '6343:6343/udp'  
- '2055:2055/udp'  
  
networks:  
- final_project_default
```

Использовалась стандартная конфигурация goflow, с дополнительными параметрами -kafka.topic flows
-proto.fixedlen=true

6 Настройка Kafka

```
kafka:
image: docker.io/bitnami/kafka:4.0
ports:
- "9092:9092"
environment:
# KRaft settings
- KAFKA_CFG_NODE_ID=0
- KAFKA_CFG_PROCESS_ROLES=controller,broker
- KAFKA_CFG_CONTROLLER_QUORUM_VOTERS=0@kafka:9093
# Listeners
- KAFKA_CFG_LISTENERS=PLAINTEXT://:9092,CONTROLLER://:9093
- KAFKA_CFG_ADVERTISED_LISTENERS=PLAINTEXT://:9092
- KAFKA_CFG_LISTENER_SECURITY_PROTOCOL_MAP=CONTROLLER:PLAINTEXT,PLAINTEXT:PLAINTEXT
- KAFKA_CFG_CONTROLLER_LISTENER_NAMES=CONTROLLER
- KAFKA_CFG_INTER_BROKER_LISTENER_NAME=PLAINTEXT
networks:
- final_project_default
```

Использовалась стандартная конфигурация goflow, с дополнительными параметрами -kafka.topic flows
-proto.fixedlen=true

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Dashboard
local-kraft
Brokers
Topics
Consumers

Topics / flows

Overview

Messages

Consumers

Settings

Statistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Submit

Oldest First

Search

+ Add Filters

DONE

53 ms

8 KB

92 messages consumed

	Offset	Partition	Timestamp	Key Preview	Value Preview
+	0	0	8/5/2025, 18:12:32		\02@ @< X:UqH...
+	1	0	8/5/2025, 18:12:32		\02@ @< X:Uq;wH...
+	2	0	8/5/2025, 18:12:32		\02@ @< X:E:wH...
+	3	0	8/5/2025, 18:12:32		\02@ @< X:E:wH...
+	4	0	8/5/2025, 18:12:32		\02@ @< X:EH...
+	5	0	8/5/2025, 18:12:32		\02@ @< X:EH...
+	6	0	8/5/2025, 18:12:32		[02@ @< X:EH...
+	7	0	8/5/2025, 18:12:32		\02@ @< X:EH...
+	8	0	8/5/2025, 18:12:32		]02@ @< X:EH...
+	9	0	8/5/2025, 18:12:32		]02@ @< X:EH...
+	10	0	8/5/2025, 18:12:32		]02@ @< X:EH...

Данные Netflow в топике Kafka

8 Настройка Kafka

Dashboard

local-kraft • ^

Brokers

Topics

Consumers

Produce Message

Topics / flows

Overview Messages Consumers Settings Statistics

Seek Type Partitions Key Serde Value Serde

Offset Offset All items are selected. String String Clear all Submit Oldest First

Q Search + Add Filters

DONE 53 ms 8 KB 92 messages consumed

	Offset	Partition	Timestamp	Key Preview	Value Preview
+	0	0	8/5/2025, 18:12:32		\o2@><X>UqH...
+	1	0	8/5/2025, 18:12:32		\o2@><Uq;q>wH...
+	2	0	8/5/2025, 18:12:32		\o2@><E>wH...
+	3	0	8/5/2025, 18:12:32		\o2@><E>wH...
+	4	0	8/5/2025, 18:12:32		\o2@><X>EH...
+	5	0	8/5/2025, 18:12:32		\o2@><X>EH...
+	6	0	8/5/2025, 18:12:32		[o2@><X>EH... ⋮
+	7	0	8/5/2025, 18:12:32		\o2@><X>EH...
+	8	0	8/5/2025, 18:12:32		]o2@><X>EH...
+	9	0	8/5/2025, 18:12:32		]o2@><X>EH...
+	10	0	8/5/2025, 18:12:32		]o2@><X>EH...

Данные Netflow в топике Kafka

9 Настройка Clickhouse

```
CREATE TABLE IF NOT EXISTS flows
(
    TimeReceived UInt64,
    TimeFlowStart UInt64,

    SequenceNum UInt32,
    SamplingRate UInt64,
    SamplerAddress FixedString(16),

    SrcAddr FixedString(16),
    DstAddr FixedString(16),

    SrcAS UInt32,
    DstAS UInt32,

    EType UInt32,
    Proto UInt32,

    SrcPort UInt32,
    DstPort UInt32,

    Bytes UInt64,
    Packets UInt64
) ENGINE = Kafka()
SETTINGS
    kafka_broker_list = 'kafka:9092',
    kafka_topic_list = 'flows',
    kafka_group_name = 'clickhouse',
    kafka_format = 'Protobuf',
    kafka_schema = 'flow.proto:FlowMessage';
```

```
CREATE TABLE IF NOT EXISTS flows_raw
(
    Date Date,
    TimeReceived DateTime,
    TimeFlowStart DateTime,

    SequenceNum UInt32,
    SamplingRate UInt64,
    SamplerAddress FixedString(16),

    SrcAddr FixedString(16),
    DstAddr FixedString(16),

    SrcAS UInt32,
    DstAS UInt32,

    EType UInt32,
    Proto UInt32,

    SrcPort UInt32,
    DstPort UInt32,

    Bytes UInt64,
    Packets UInt64
) ENGINE = MergeTree()
PARTITION BY Date
ORDER BY TimeReceived;
```

```
CREATE MATERIALIZED VIEW IF NOT EXISTS flows_raw_view TO flows_raw
AS SELECT
    toDate(TimeReceived) AS Date,
    *
    FROM flows;

CREATE TABLE IF NOT EXISTS flows_5m
(
    Date Date,
    Timeslot DateTime,

    SrcAS UInt32,
    DstAS UInt32,

    ETypeMap Nested (
        EType UInt32,
        Bytes UInt64,
        Packets UInt64,
        Count UInt64
    ),

    Bytes UInt64,
    Packets UInt64,
    Count UInt64
) ENGINE = SummingMergeTree()
PARTITION BY Date
ORDER BY (Date, Timeslot, SrcAS, DstAS, \`ETypeMap.EType\`);
```

Настройка схемы базы данных

10 Настройка Apache Superset

```
I have no name!@25dad855d1b:/opt/bitnami/airflow$ airflow users create --username admin --password admin --role Admin -e 1@1.ru -f admin -l admin
/opt/bitnami/airflow/venv/lib/python3.12/site-packages/flask_limiter/extension.py:324 UserWarning: Using the in-memory storage for tracking rate limits as
no storage was explicitly specified. This is not recommended for production use. See: https://flask-limiter.readthedocs.io#configuring-a-storage-backend
for documentation about configuring the storage backend.
[2025-08-05T16:31:11.493+0000] {override.py:1362} INFO - Added user admin
User "admin" created with role "Admin"
I have no name!@25dad855d1b:/opt/bitnami/airflow$
```

Connect a database

STEP 2 OF 3

Enter the required ClickHouse Connect (Superset) credentials

Need help? Learn more about [connecting to ClickHouse Connect \(Superset\)](#).

Host * Port

Database name

Copy the name of the database you are trying to connect to.

Username

Password

Display Name *

Pick a nickname for how the database will display in Superset.

Additional Parameters

Add additional custom parameters

☐ SSL ⓘ

[Connect this database with a FQDN ClickHouse URL instead. ⓘ](#)

[Back](#) [Connect](#)

Connect a database

STEP 3 OF 3

Database connected

Create a dataset to begin visualizing your data as a chart or go to SQL Lab to query your data.

[Create dataset](#) [Query data in SQL Lab](#)

SQL Lab
Adjust how this database will interact with SQL Lab.

Performance
Adjust performance settings of this database.

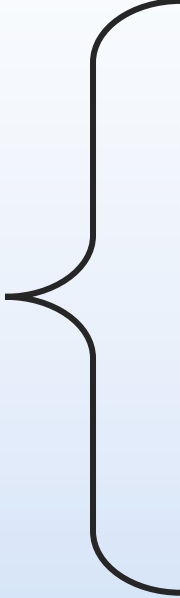
Security
Add extra connection information.

Other
Additional settings.

[Back](#) [Finish](#)

Настройка подключения к Clickhouse

ВЫВОДЫ



В работе была получена работающая связка по обработке данных из реального устройства по протоколу Netflow с использованием Clickhouse и позволяющая вести анализ в Apache Superset